Demonstration of High-Gradient in a Cryo-Cooled X-Band Structure

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Technology and Innovation Directorate, SLAC

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U.S. Department of Energy Office of Science

Stanford ENGINEERING Electrical Engineering

SLAC NATIONAL ACCELERATOR LABORATORY

Outline

- **Physics of breakdown**

  in high gradient normal conducting structures

- **The distributed-coupling technology**

  New geometric optimization capabilities

- **Cryogenic-copper structures**

  New accelerating frontiers for high-gradient operation

- **Next-Generation Distributed-Coupling Linacs**

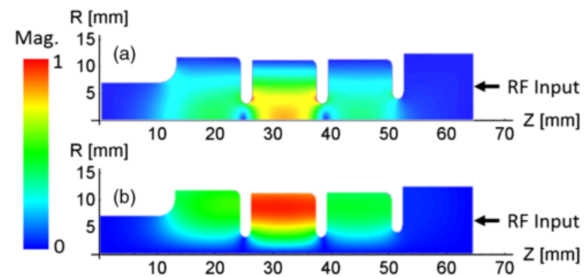
  Making full benefit of the optimization capability of this novel technology

Understanding the physics of breakdowns triggered new research directions for high gradient linear accelerators

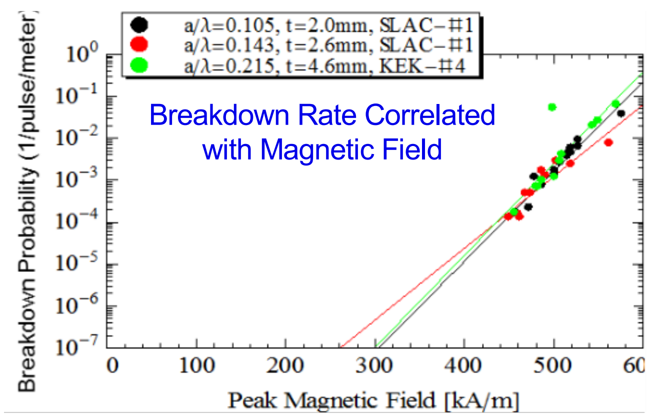
Experimental studies for understanding the physics of breakdowns have led to many discoveries

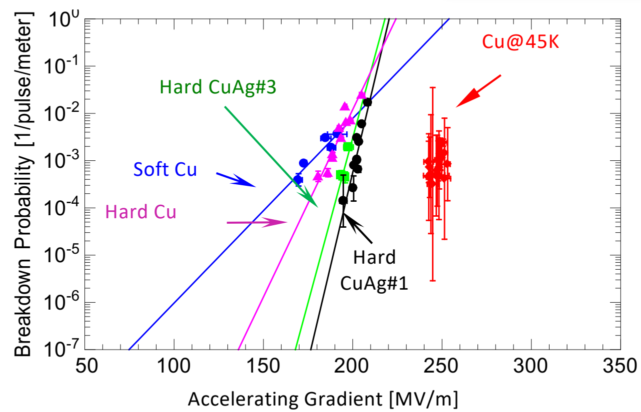
- **Magnetic field**'s role in breakdowns through pulsed heating
- Reduced breakdown rates with increased **material strength** (hard materials and low temperature operation)

Single-cell testing



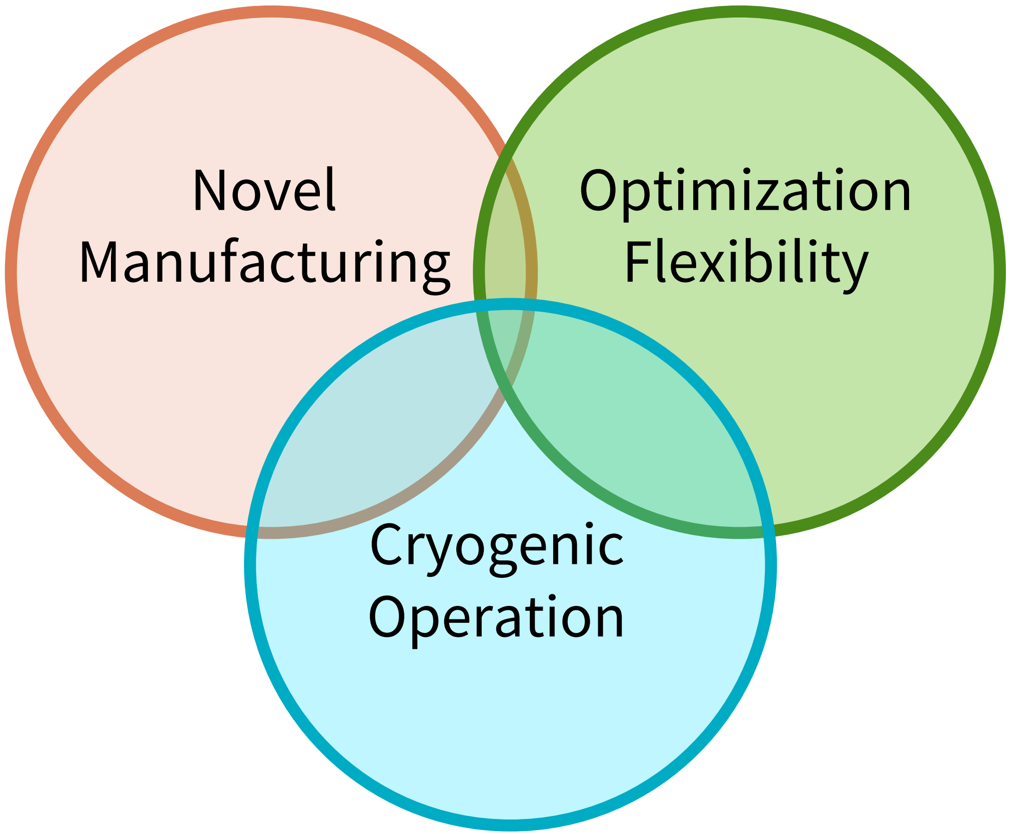
Full scale structures are long, complex, and expensive





Valery Dolgashev et al., "Geometric dependence of radio-frequency breakdown in normal conducting accelerating structures," Applied Physics Letters, Volume 97, Issue 17, pp 171501 - 171501-3, October 2010.
Alex Cahill et al., "High gradient experiments with X-band cryogenic copper accelerating cavities" Phys. Rev. Accel. Beams 21, 102002 – Published 23 October 2018.

Understanding the physics of breakdowns triggered new research directions for high gradient linear accelerators

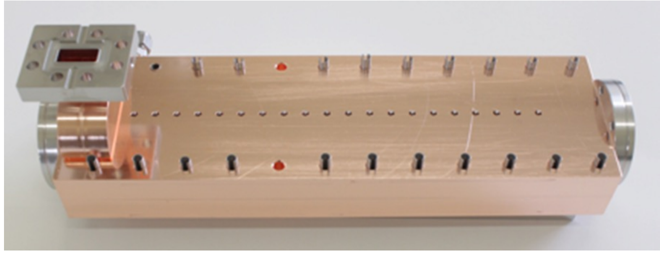


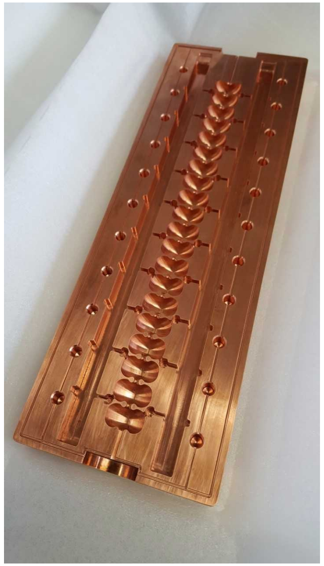
Novel Manufacturing Techniques

Novel fabrication methods are compatible with hard metals and reduced cost

- Milling structure out of two halves instead of machining and then brazing single cells
- No RF currents through the joint
- Consistent with manufacturing structures out of **hard copper alloys**

Demonstrated with Standing and Traveling Wave Structure





S. Tantawi, P. Borchard, Z. Li
V. Dolgashev

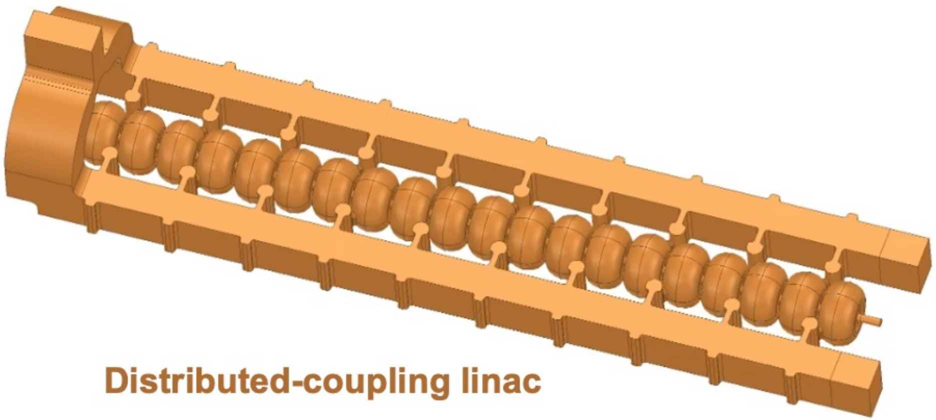
Developing new manufacturing techniques to implement advanced materials in practical accelerating structures and reduce costs

Optimized Geometries using Distributed-Coupling Topology

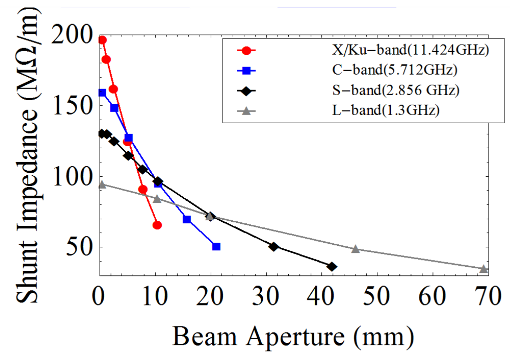
Distributed coupling Linacs provides a new technology that enables much flexible cell-design optimization

The existing linac designs requires careful consideration of the coupling between cells which limits the ability of designers to optimize the cell shape.

The distributed coupling linac overcomes these limitations by providing a new topology that allows feeding each accelerator cell independently using a periodic feeding network.



Distributed-coupling linac



New Scaling Laws Determine the Best Performance for Accelerating Structures

Tantawi et al. "Distributed coupling and multi-frequency microwave accelerators," Jul. 5 2016, US Patent 9,386,682.

S. Tantawi, M. Nasr et. al. "Design and demonstration of a distributed-coupling linear accelerator structure." *Physical Review Accelerators and Beams 23.9 (2020): 092001.*

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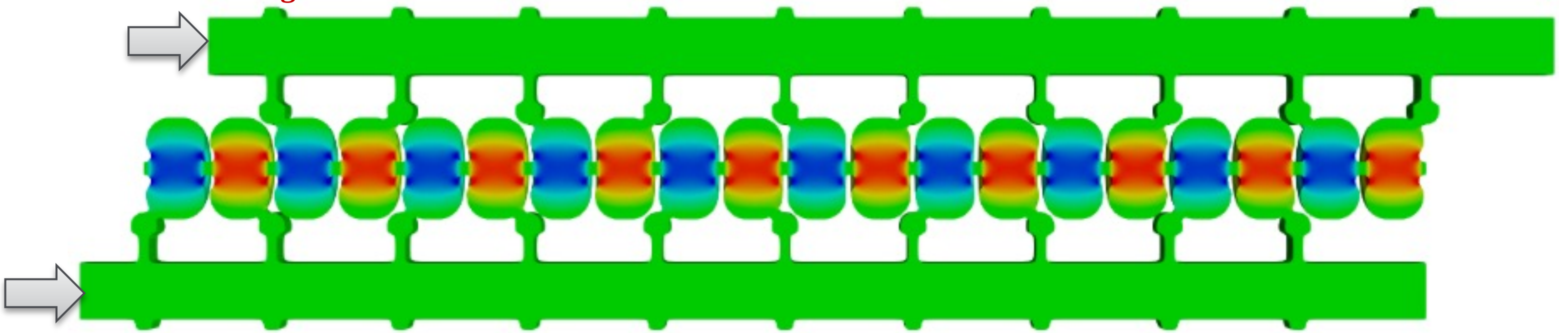
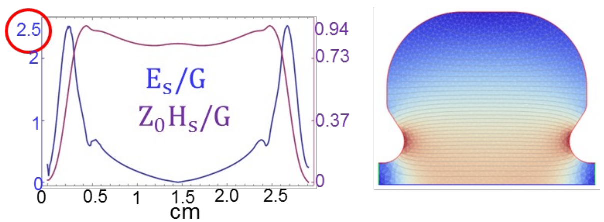
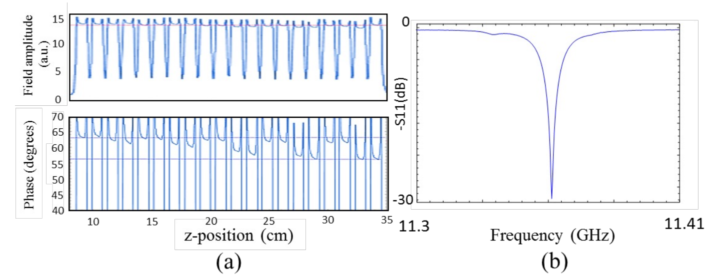


Designed to minimize surface magnetic field

π-mode, X-band, 20 cells Linac $R_{shunt} = 155\ \mathrm{M\Omega/m}$



Easy tuning and single frequency rather than 20



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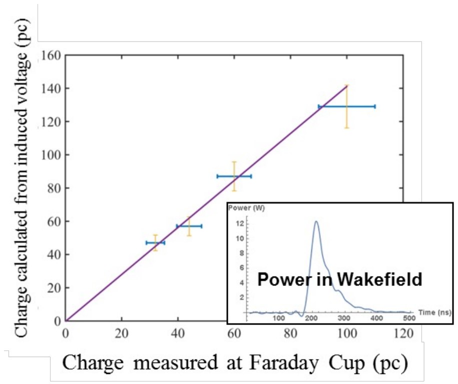
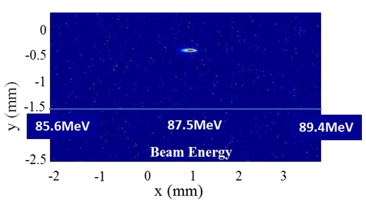
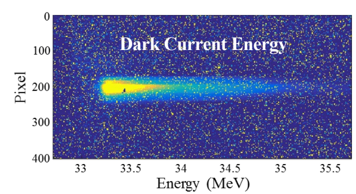
Verification of accelerating parameters and study of breakdown mechanism in the first distributed-coupling linac

Verification using beam energy, dark current, and charge measurements



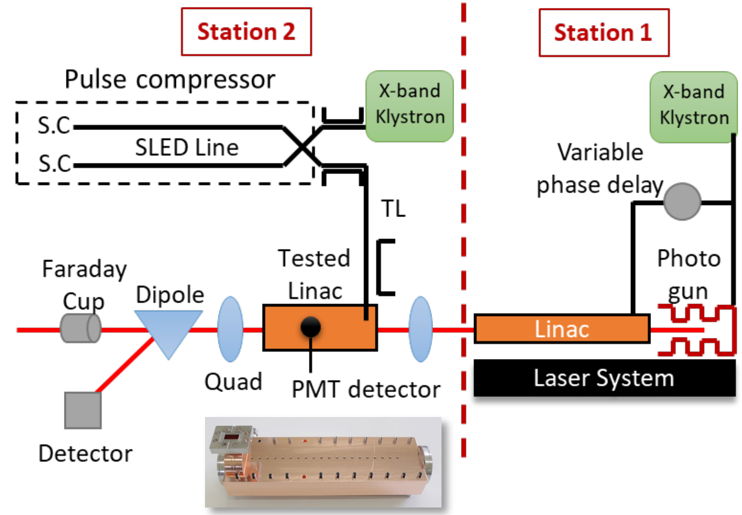


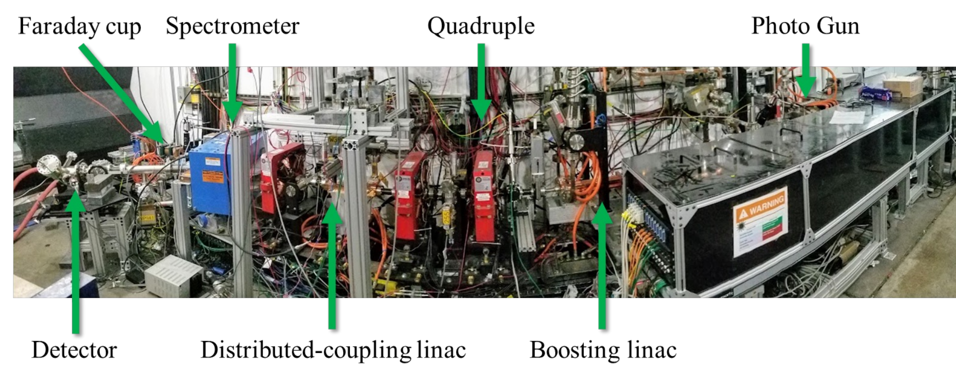
Gradient=100 MV/m



Gradient=140 MV/m

Results agree with the designed shunt impedance and predicted quality factor





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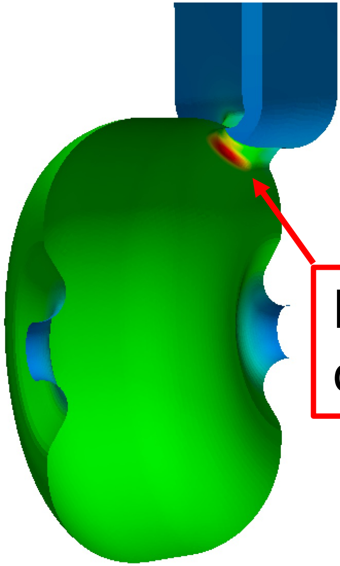
Verification of accelerating parameters and study of breakdown mechanism in the first distributed-coupling linac

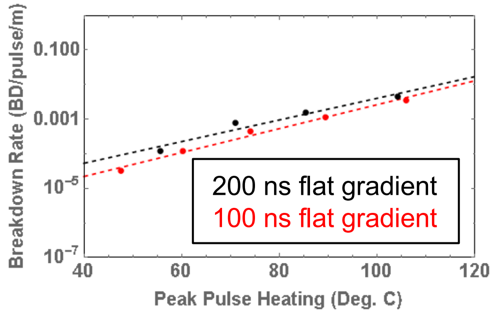
$\text{Peak } E_s/\text{Gradient} = 2.5$



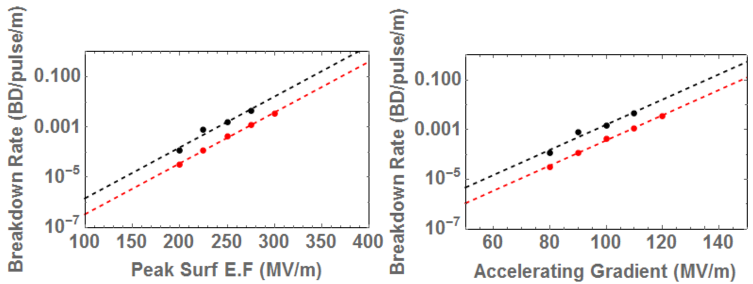
H field enhancement at coupler iris: ×2.28

H Field

Breakdown measurements at 100 ns and 200 ns flat gradient.



Pulse heating lines matches for both pulses



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Cryogenic Operation
of
Normal Conducting Structures

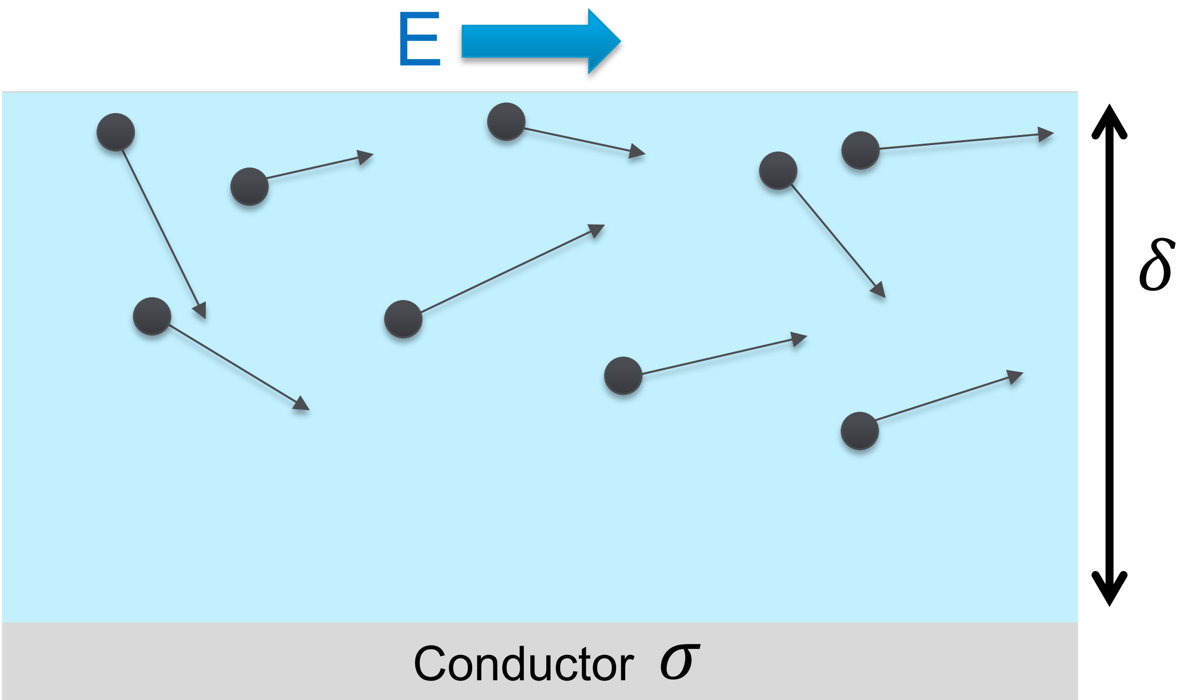
Cryogenic operation of NC accelerating structures

- **Increase conductivity**
  - **Low losses** allows for heavy beam loading
  - **Enhanced shunt impedance** increases efficiency

- **Increase material strength reduces breakdown rates**

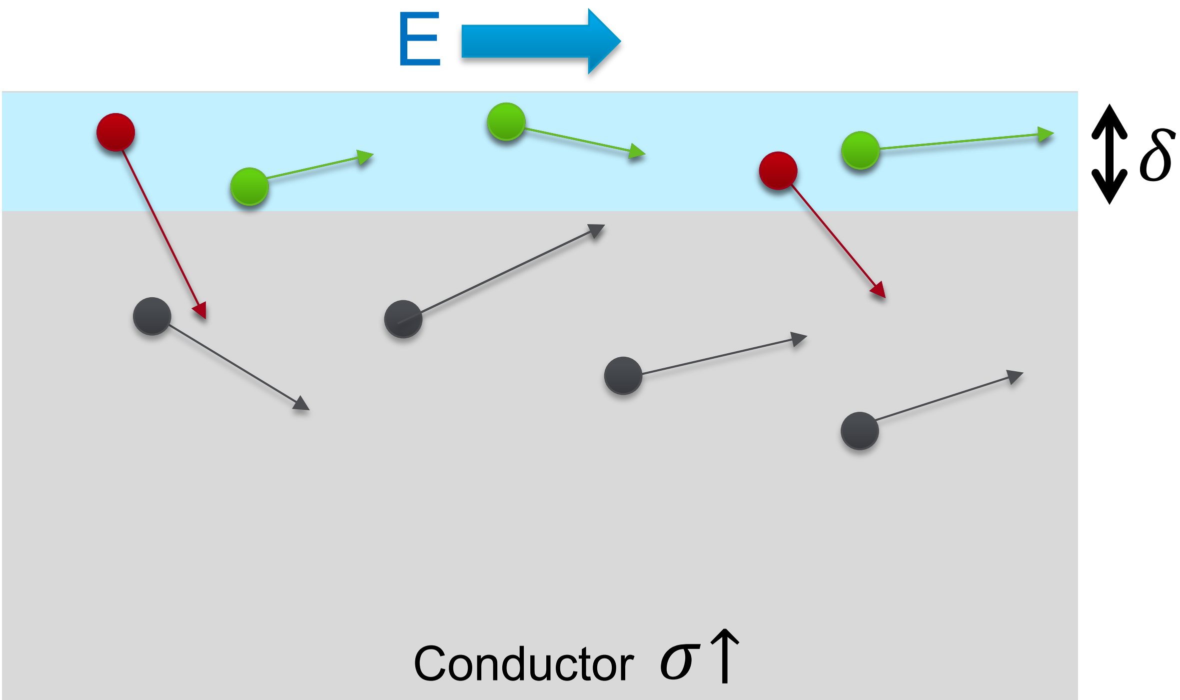
The theory of Anomalous skin effect in metals

At high temperatures and low frequencies, the mean free path of the electrons in a good conductor is much smaller than the classical skin depth.



The theory of Anomalous skin effect in metals

At low temperatures and high frequencies, the mean free path of the electrons in a good conductor becomes greater than the classical skin depth.



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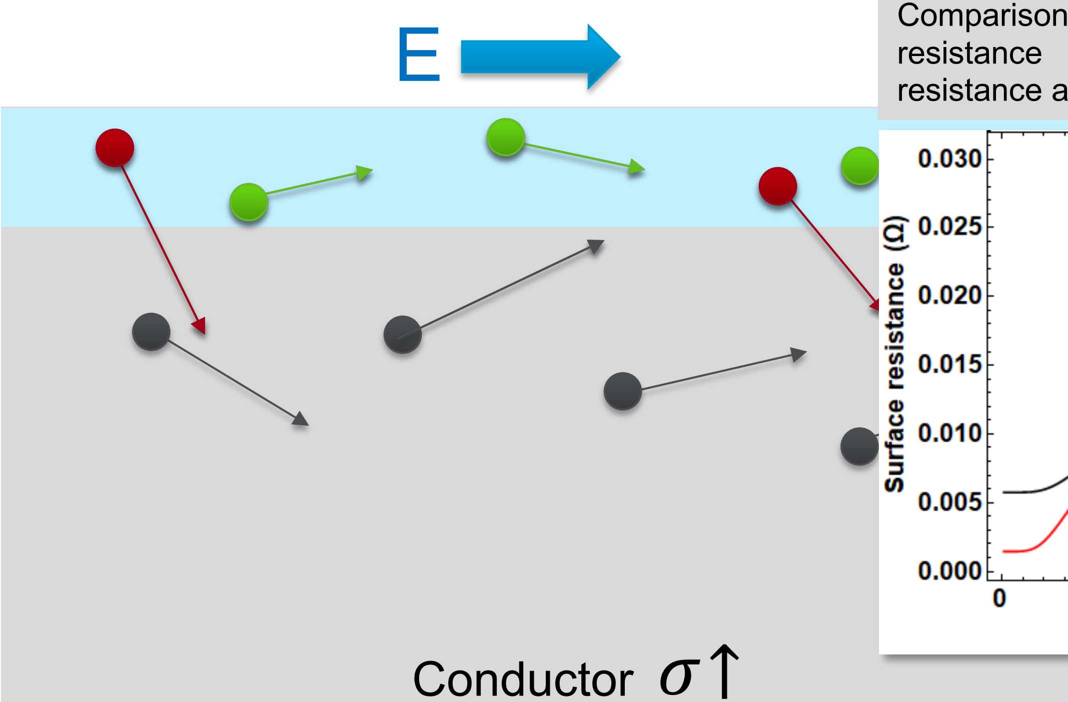
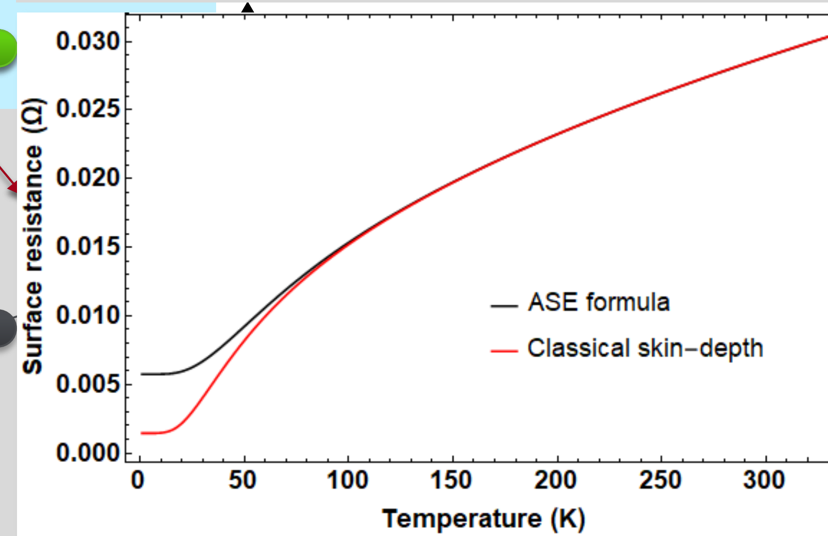
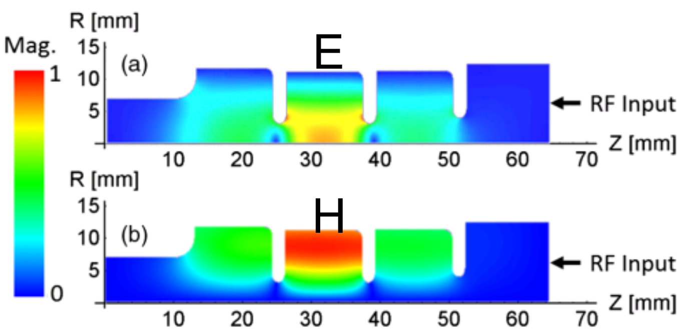


Comparison of the normal skin effect surface resistance and anomalous skin effect surface resistance at 11.424 GHz in RRR=400 copper

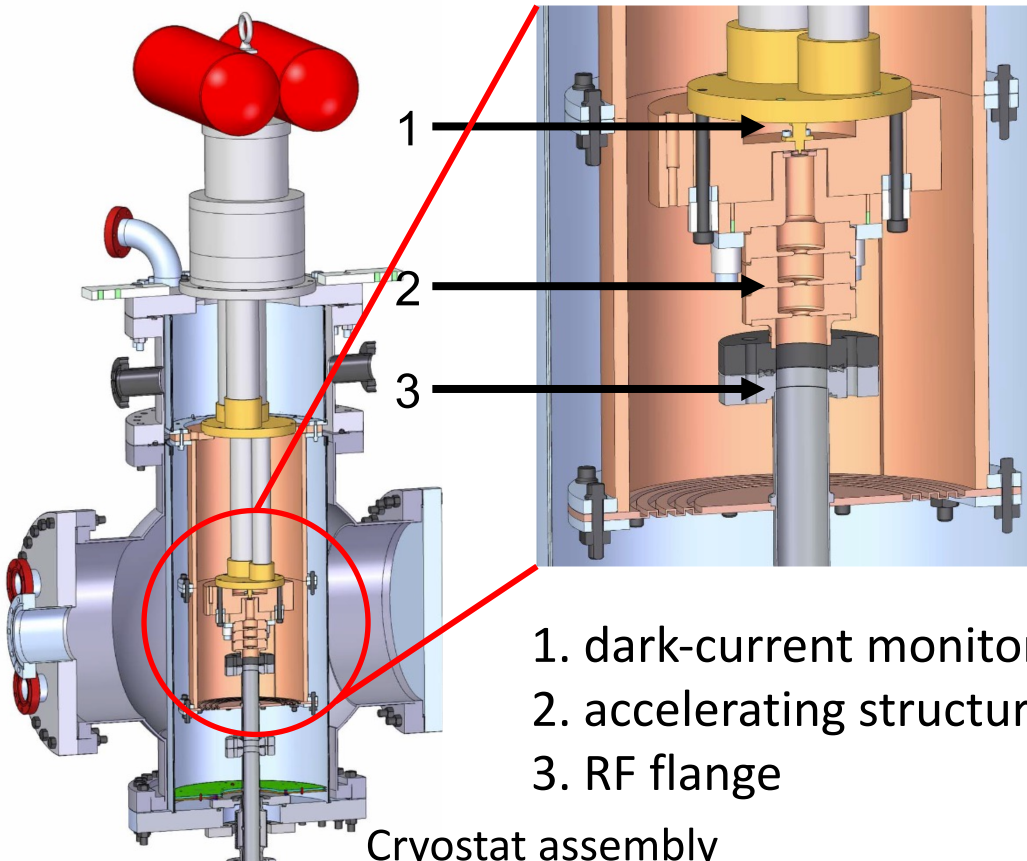


High-power testing of single-cell structure at cryogenic temperature of 45k

Single-cell testing



The single-cell fields profile



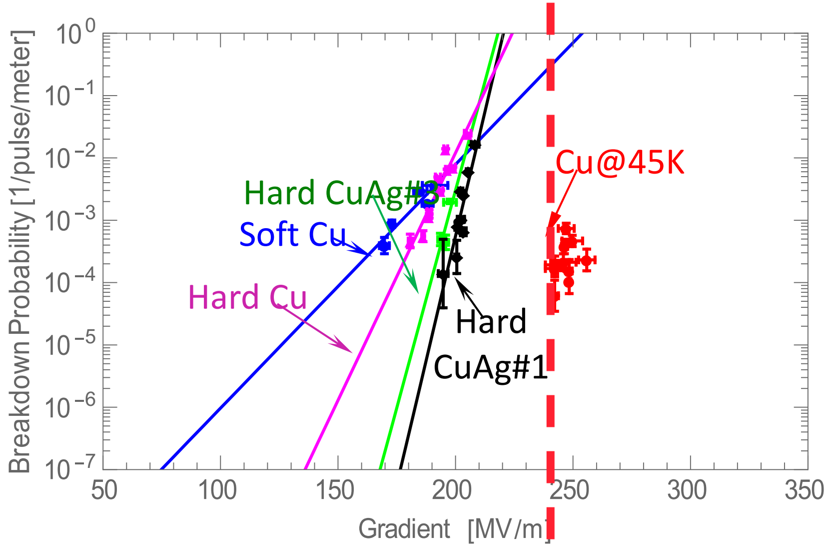
1. dark-current monitor
2. accelerating structure
3. RF flange

Cryostat assembly

A. Cahill *et al*., "High gradient experiments with X-band cryogenic copper accelerating cavities," PRAB, 21, 2018

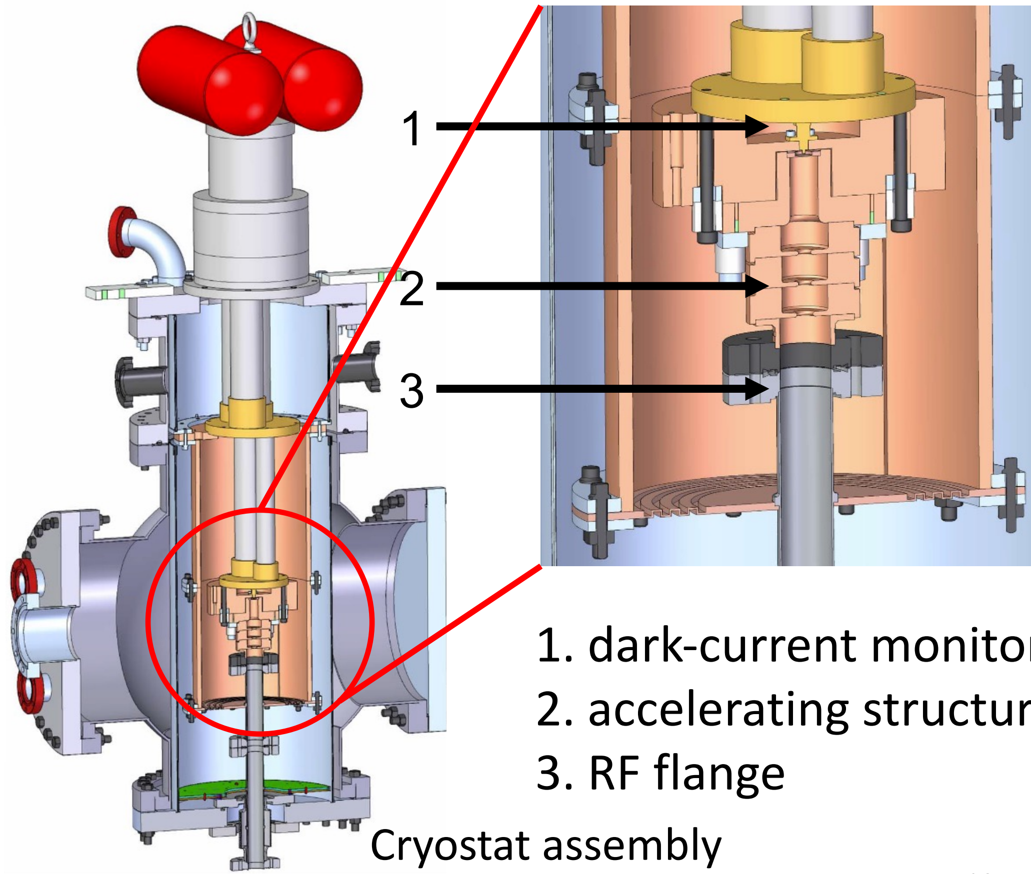
High-power testing of single-cell structure
at cryogenic temperature of 45k

Breakdowns were detected above a gradient of 240 MV/m



Threshold

A. Cahill *et al.*, "High gradient experiments with X-band cryogenic copper accelerating cavities," PRAB, 21, 2018



1. dark-current monitor
2. accelerating structure
3. RF flange

Cryostat assembly

Full practical accelerator testing at cryogenic temperature

- **Distributed-coupling technology**

  with flexible optimization and exceptional Rs

- **Cryogenic operation**

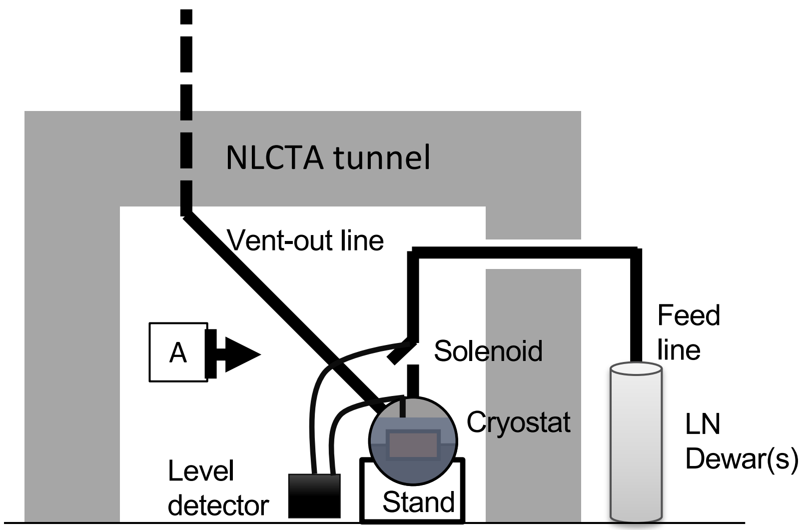
  for increased Rs and reduced breakdown rates

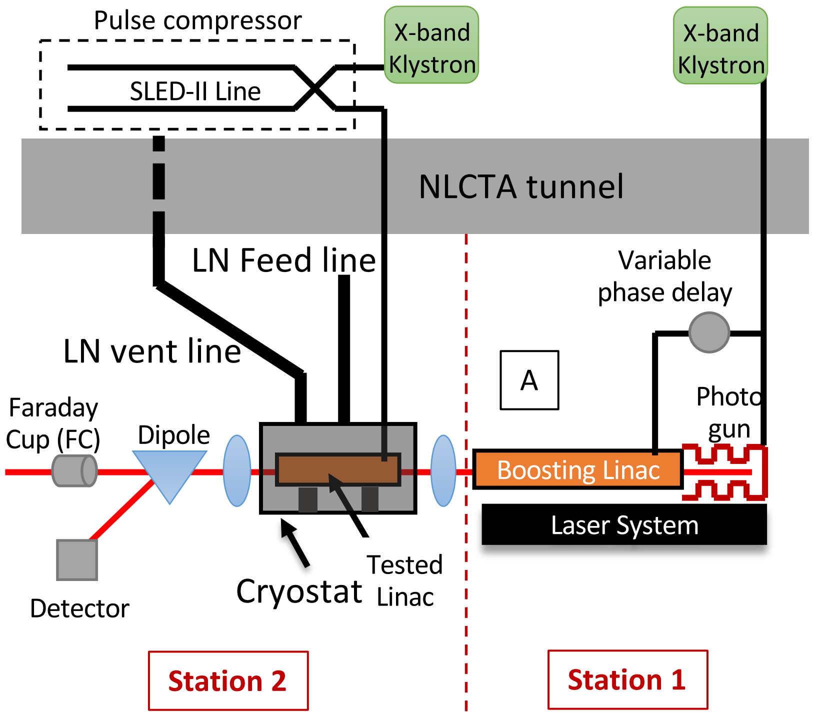
- **Dramatic reduction in cost**

  using cryogenics at 77K

High-power test of the distributed-coupling structure (Xband, 20 cells) at LN cryogenic temperature (77k)

The high-power experiment was modified for LN cryogenic operation





M. Nasr, S. Tantawi, E. Nanni et. al.

Dramatic reduction in cost of system including cryogenics at 77K

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SLAC

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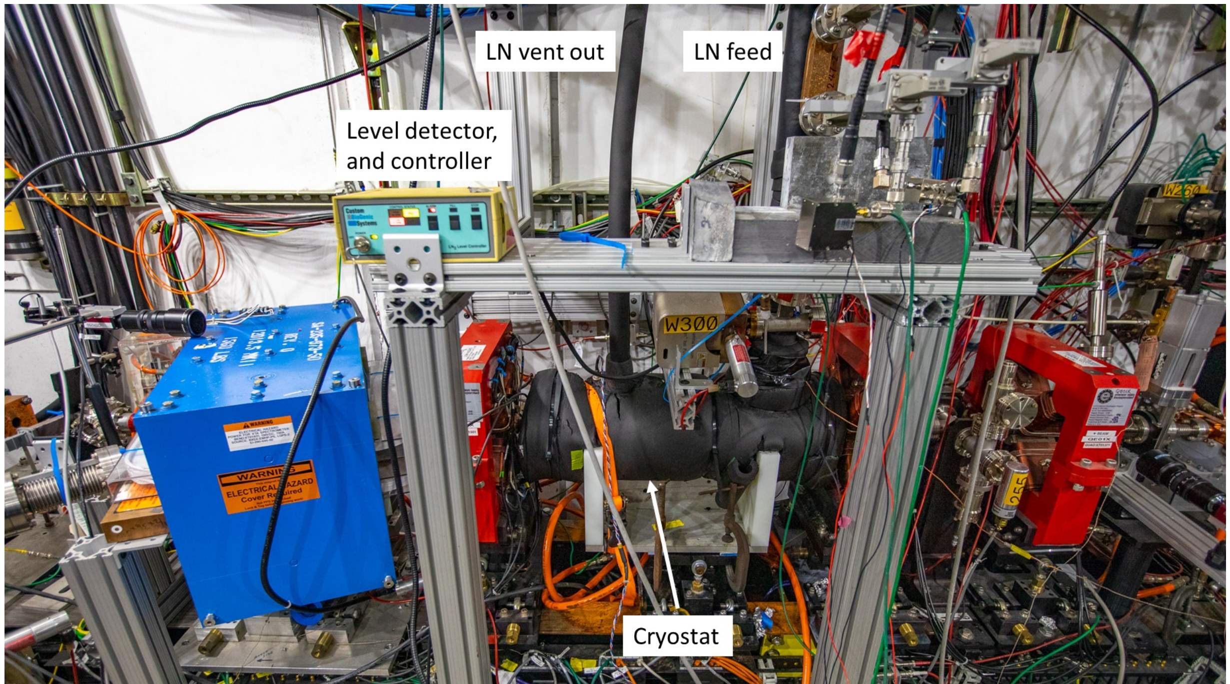

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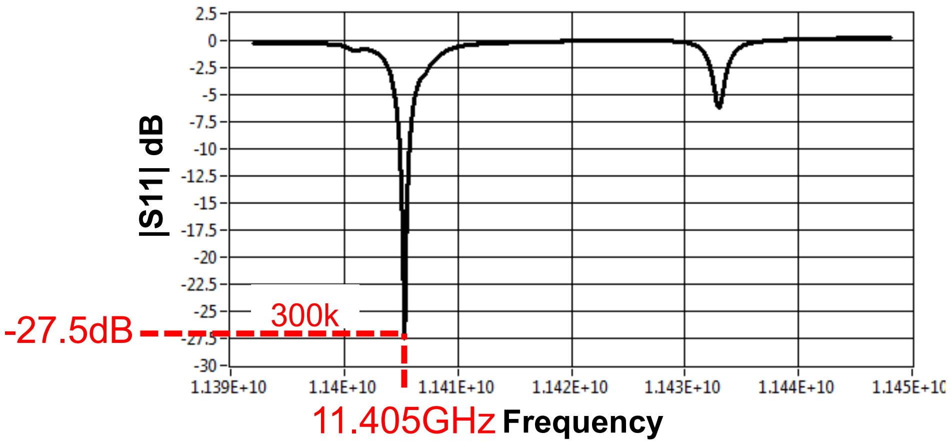
Cold cold-test of the distributed-coupling structure (Xband, 20 cells Linac) at LN cryogenic temperature (77k)



M. Nasr et.al. *"Experimental demonstration of particle acceleration with normal conducting accelerating structure at cryogenic temperature,"* arXiv preprint arXiv:2011.00391

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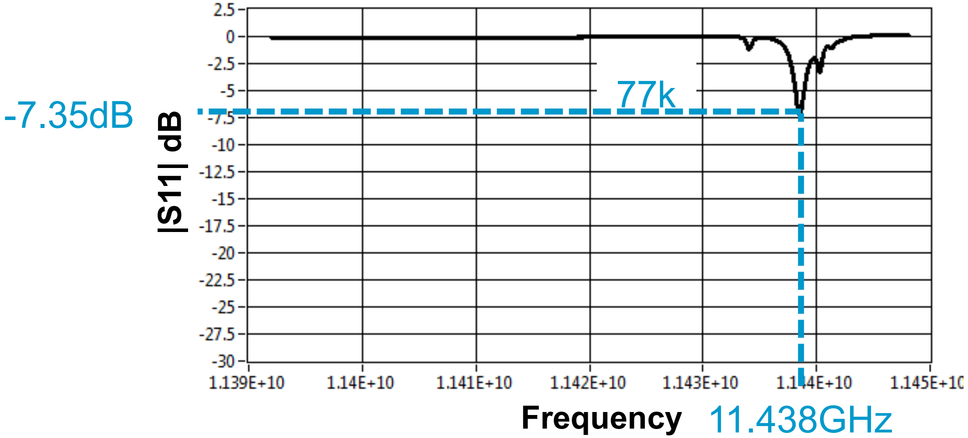

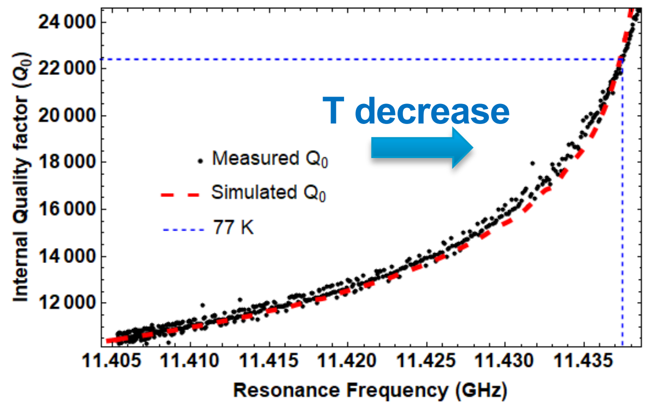

2.25X less RF power loss

Shunt impedance of 350 MΩ/m

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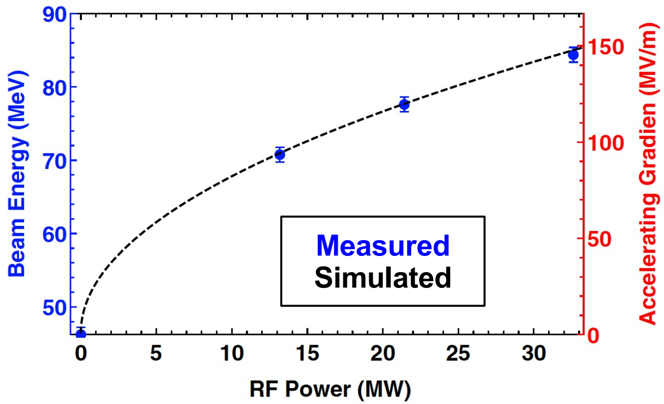
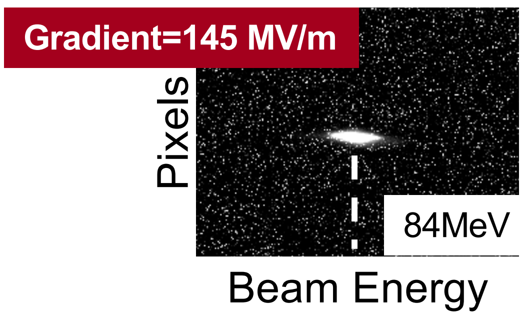
High-power test of the distributed-coupling structure (Xband, 20 cells) at LN cryogenic temperature (77k)

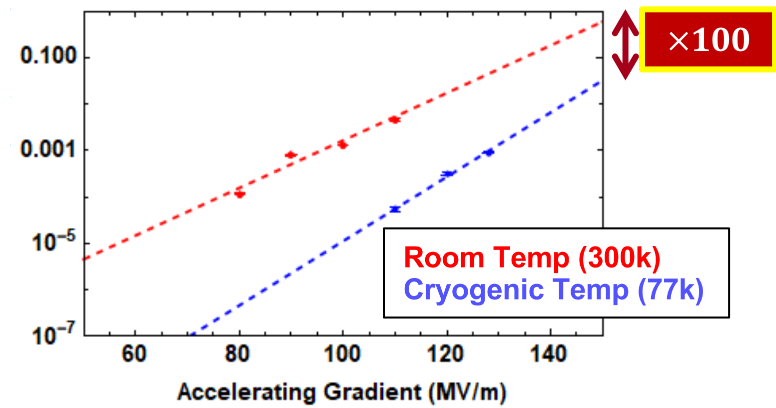
SLAC

Modeled structure gradient confirmed with electron beam





Reduced breakdown rates compared to 300k operation



Stepped pulse with flat gradient of 200ns

M. Nasr et al. "Experimental demonstration of particle acceleration with normal conducting accelerating structure at cryogenic temperature." *Physical Review Accelerators and Beams 24.9 (2021): 093201.*

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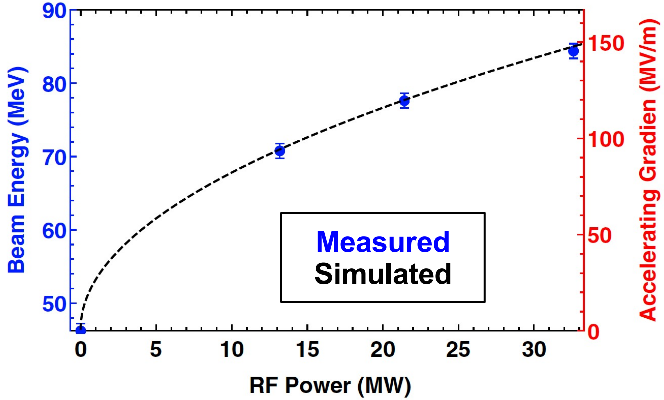

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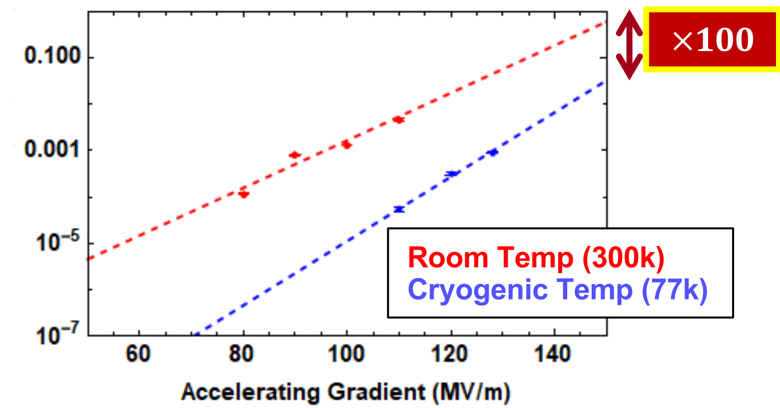


Stepped pulse with flat gradient of 200ns

	300k	**77k**
$\beta = Q_0/Q_e$	1	2.25
R_s (MΩ/m)	155	350
BDR	x_0	$x_0/100$

- Heavy beam loading
- Reduced breakdown rates
- High RF-to-beam efficiency
- Cost-efficient solution with LN

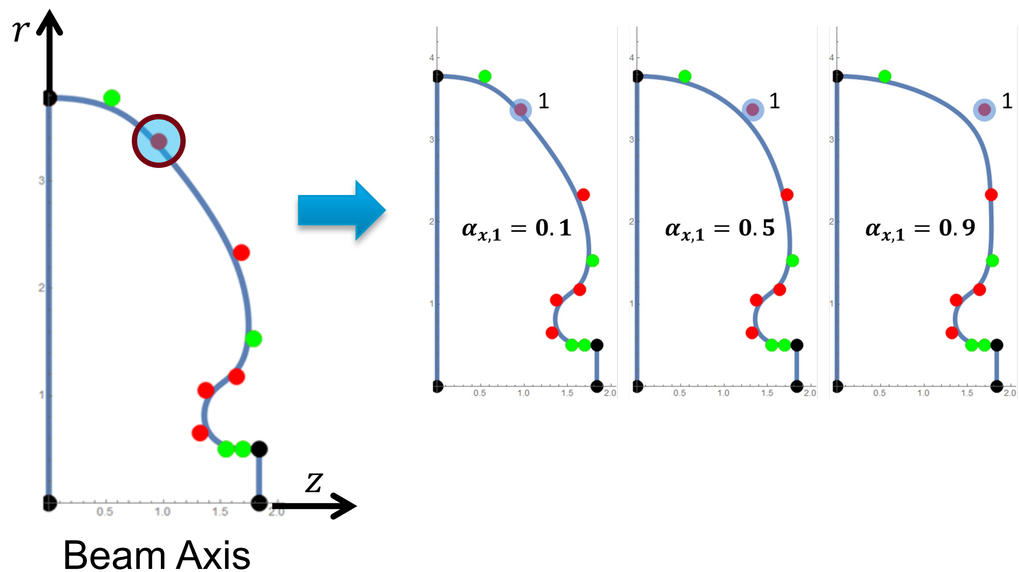
M. Nasr et al. "Experimental demonstration of particle acceleration with normal conducting accelerating structure at cryogenic temperature." *Physical Review Accelerators and Beams 24.9 (2021): 093201.*

Next-Generation
Distributed-Coupling Linac

Next-generation distributed coupling linac:
New geometric-optimization approaches and optimized ports

New geometric-optimization approaches

The cavity curvature is defined with a set of control points with variable positions to control splines which describes the cavity shape.



M. Nasr et al. New Geometrical-Optimization Approach using Splines for Enhanced Accelerator Cavities' Performance, IPAC'18

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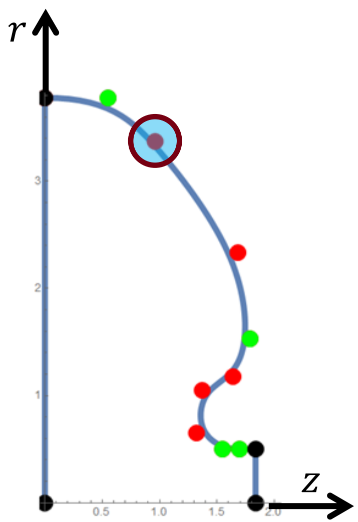


Beam Axis

M. Nasr et al. New Geometrical-Optimization Approach using Splines for Enhanced Accelerator Cavities' Performance, IPAC'18

The first design with circular shapes

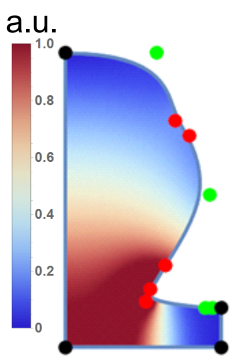
$$\mathrm{R_s} = 155\ \mathrm{M\Omega/m} \rightarrow \frac{E_{peak}}{E_{acc}} = 2.5$$

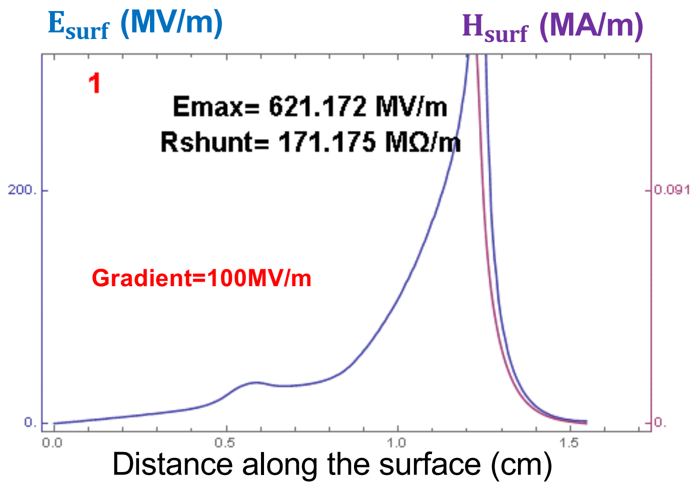
Can we do better?

New geometric-optimization approaches

X-band, 11.424 GHz, iris diameter 0.1λ

Starting shape





Target: Maximize shunt impedance with peak surface E-field to gradient ratio of 2

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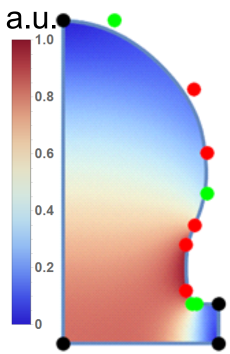
M. Nasr, Z. Li, and S. Tantawi

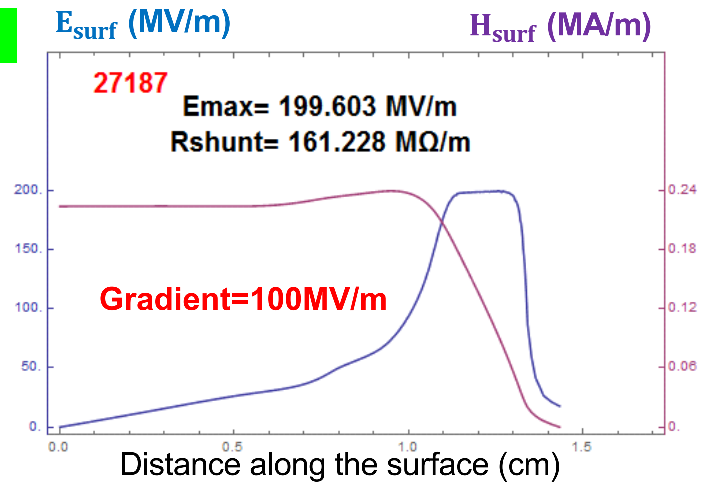
Next-generation distributed coupling linac:
New geometric-optimization approaches and optimized ports

New geometric-optimization approaches

X-band, 11.424 GHz, iris diameter 0.1λ

Optimized shape





Peak surface E-field to gradient reduction from 2.5 to 2
with even higher shunt impedance

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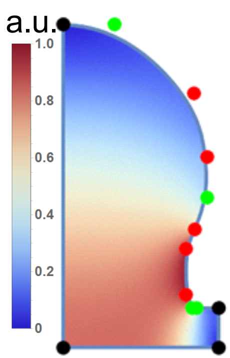
Next-generation distributed coupling linac:
New geometric-optimization approaches and optimized ports

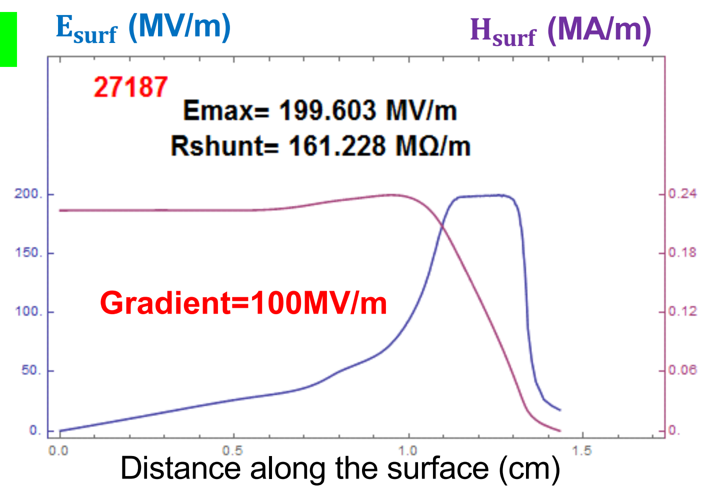
SLAC

New geometric-optimization approaches

X-band, 11.424 GHz, iris diameter 0.1λ

Optimized shape





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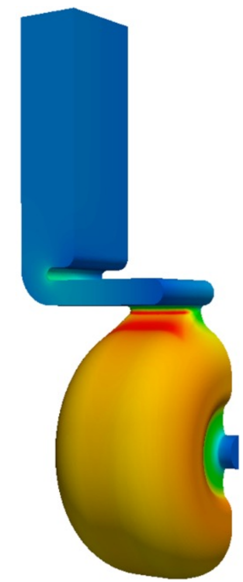
Optimized coupling ports



H Field

Reducing magnetic field enhancement to from $\times 2.3$ to $\times 1.17$

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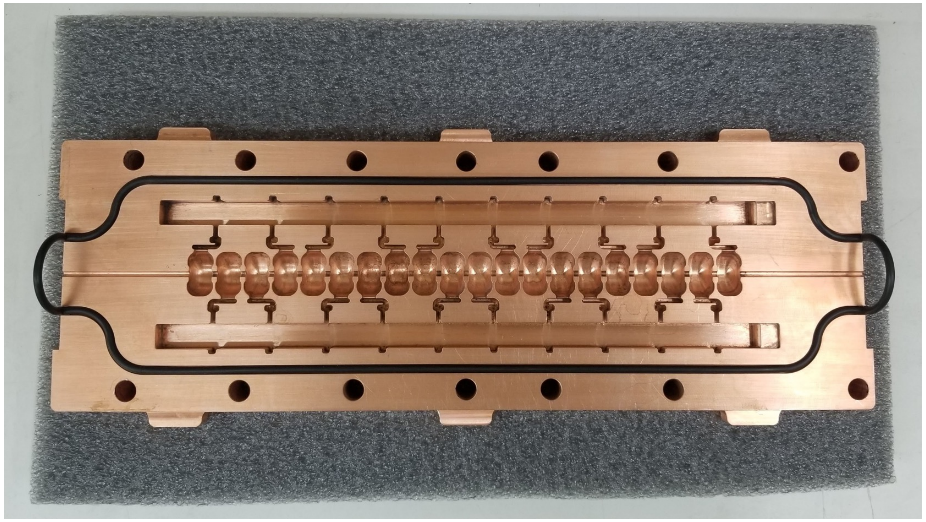
Next-generation distributed coupling linac:
New geometric-optimization approaches and optimized ports

New geometric-optimization approaches



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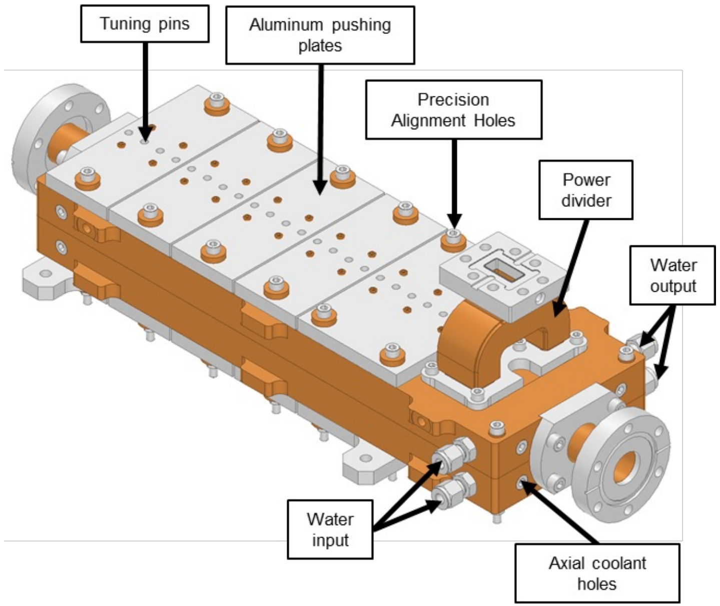


Reducing magnetic field enhancement to from ×2.3 to ×1.17

M. Nasr, Z. Li, G. Bowden, and S. Tantawi

Dual-Mode Dual-Frequency Linac

Having every cell fed individually by a manifold naturally inspires the use of another manifold with another mode feeding the same cell

SLAC

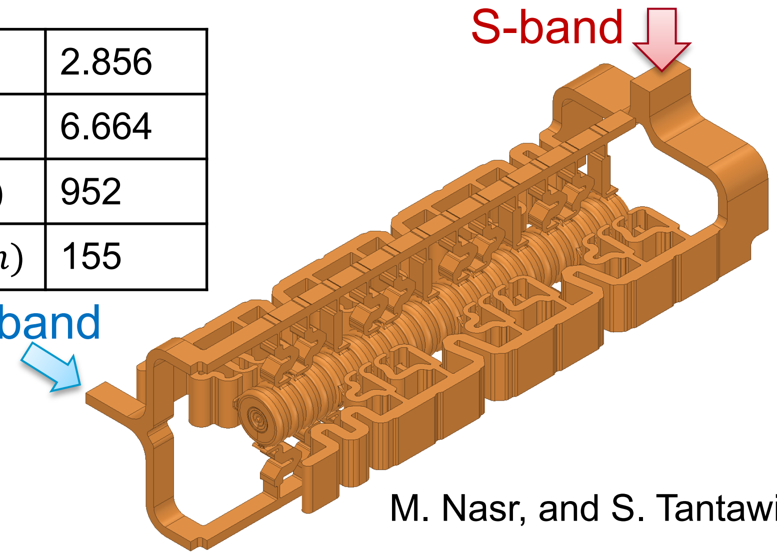
Power ∝ (gradient)²

Performance of cavity improves when powered with two different RF modes

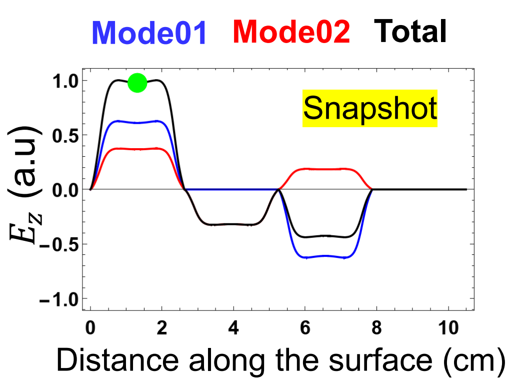
- **Efficiency:** Linear superposition of fields by adding power in the two modes.
- **Gradient:** doubling the accelerating gradient without doubling surface fields

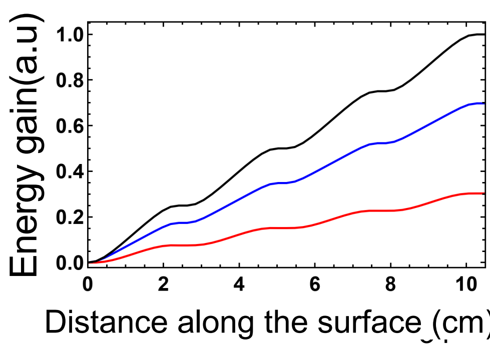
Optimum designs require non-harmonic operation; instead, we design for frequencies that are not necessarily harmonically related, but rather have a common sub-harmonic.

f_{res1}(GHz)	2.856
f_{res2}(GHz)	6.664
f_{comm}(MHz)	952
R_{shunt}(MΩ/m)	155



M. Nasr, and S. Tantawi



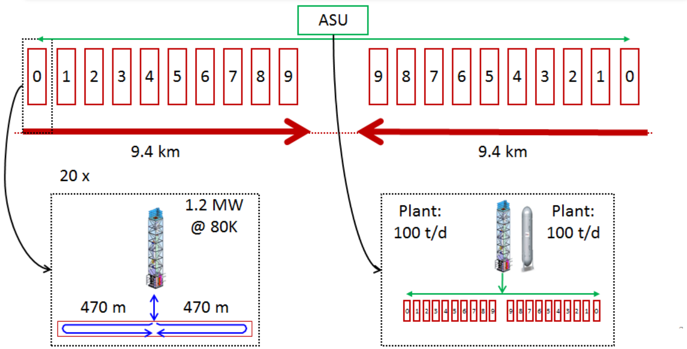


M. Nasr et al. "A Novel Dual-Mode Dual-Frequency Linac Design", IPAC'17

M. Nasr et al. "The Design and Construction of a Novel Dual-Mode Dual-Frequency Linac Design", IPAC'18

Future work and collaborations

C^3 An Advanced NCRF Linac Concept for a High Energy e+e- Linear Collider



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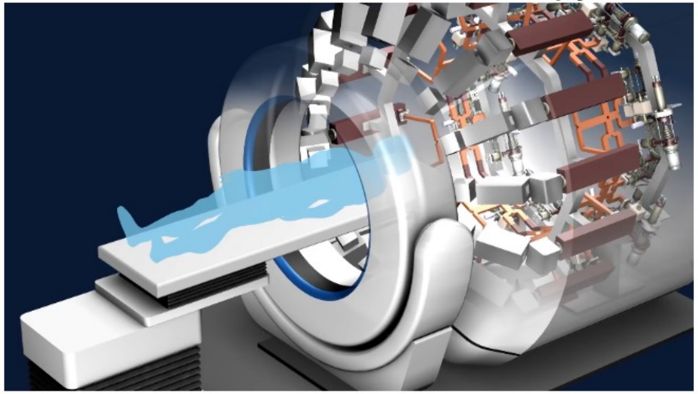

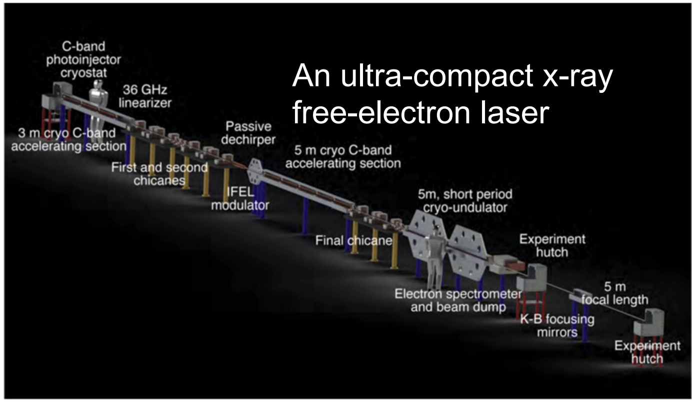

Conclusion

- **Understanding the physics of breakdowns**
  triggered new research directions for high-gradient operation

- **Distributed-coupling technology**
  enables much flexible cell-design optimization and pushes the limitation of Linacs performance.

- **Cryogenic-copper accelerating structures**
  have the promise for a new frontier for beam brightness, efficiency and cost-capability

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